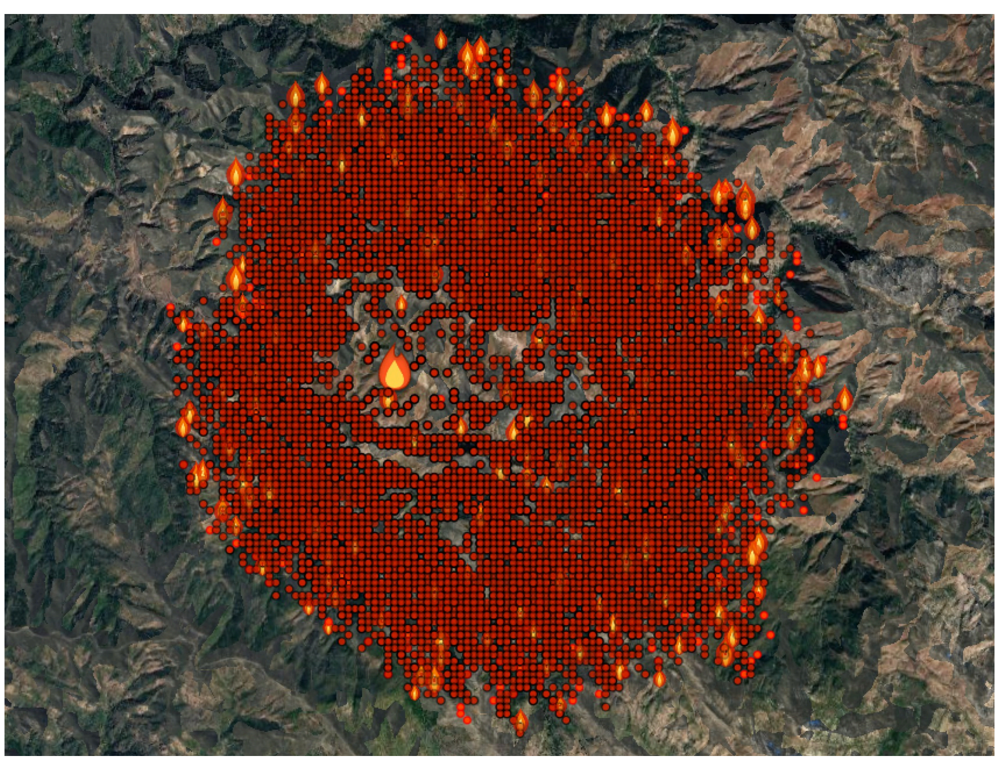
How the dog decides

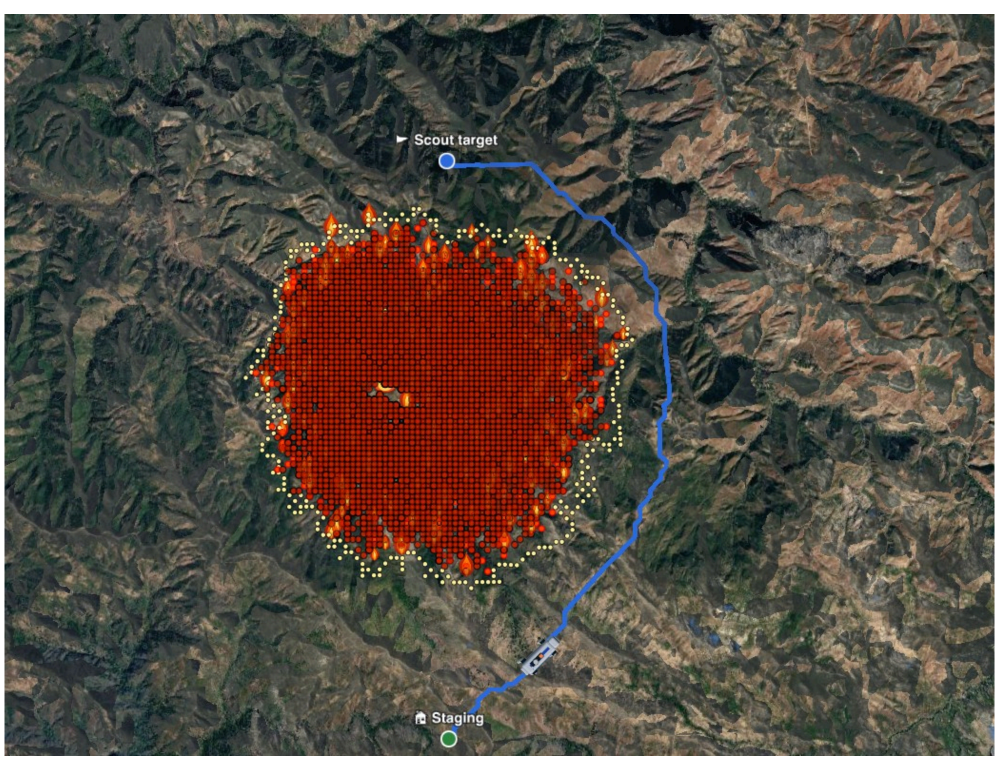
1 Predict



50 m cells **15 min** steps

Forecasts when each cell will burn.

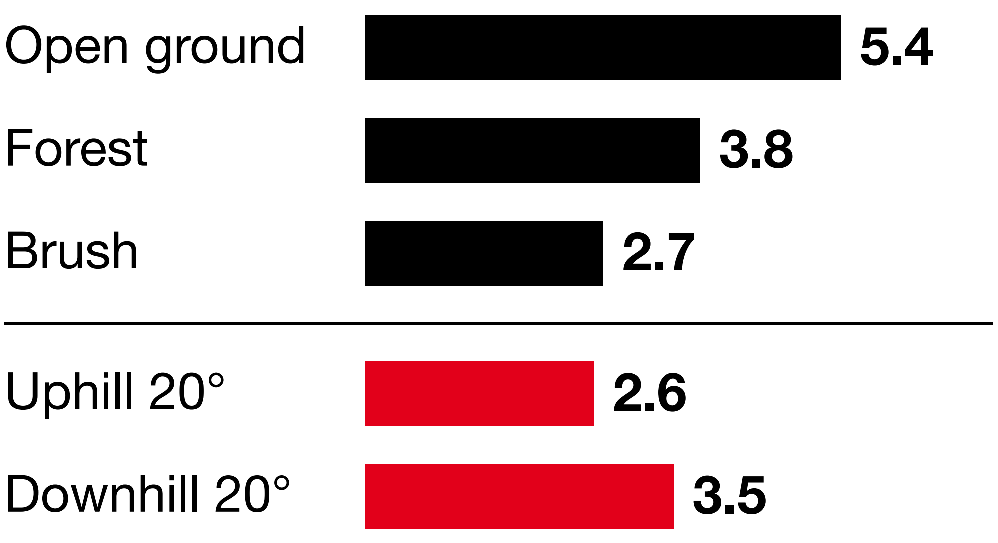
2 Safe route



Blue route to the scout target

Plans to keep 100 m from the forecast fire, with 30 min in hand.

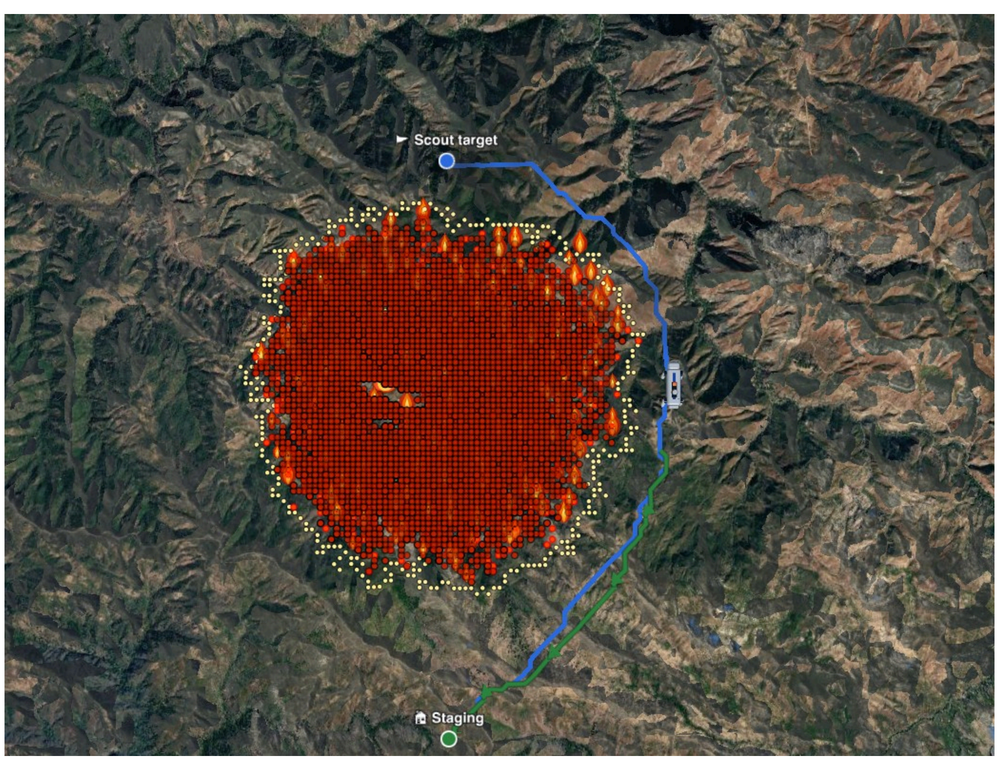
2b Route speed



Walking speed in **km/h**

The route is timed at these speeds. The planner never crosses water or slopes over 35°.

3 Escape path



Green escape path home

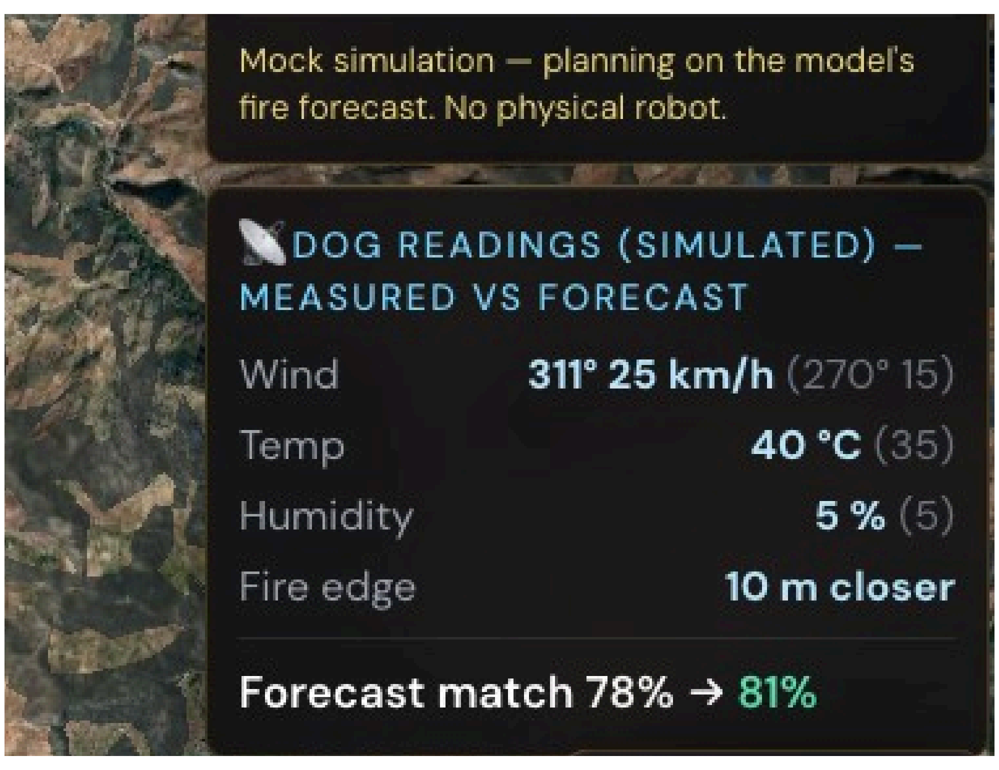
Aims to pass each point home an hour before the forecast fire comes within 100 m.

4 Scout

Wind, temperature, humidity, fire edge

Measures what the forecast only guessed.

5 Feedback



Mock simulation — planning on the model's fire forecast. No physical robot.

DOG READINGS (SIMULATED) — MEASURED VS FORECAST

Wind	**311° 25 km/h** (270° 15)
Temp	**40 °C** (35)
Humidity	**5 %** (5)
Fire edge	**10 m closer**

Forecast match 78% → 81%

Launch	Before report	After
84%	**78%**	**81%**

The readings improve the next forecast.

Simulator views on real terrain. Yellow dotted edge: the forecast the dog plans on; the real fire is hidden from it until its sensors see it. Forecast match: overlap of the forecast's and the real fire's new spread 4 h ahead, both simulated. Card numbers are planning steps, not the story's stops. Robot image: design concept; sensors are simulated.

Status and next

Done

- ML fire forecast trained and tested
- 3D browser simulator on real terrain
- Containment planner (search)
- Robot-dog loop in mock simulation

Next

- Forecast + planner on the dog's controller
- Real sensors feeding the forecast
- Scout the planned containment line
- Tests on real robot dogs

Open-source code and demo video

Data and references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections); Open-Meteo weather archive; ESA WorldCover 10 m land cover; Cesium World Terrain; Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*